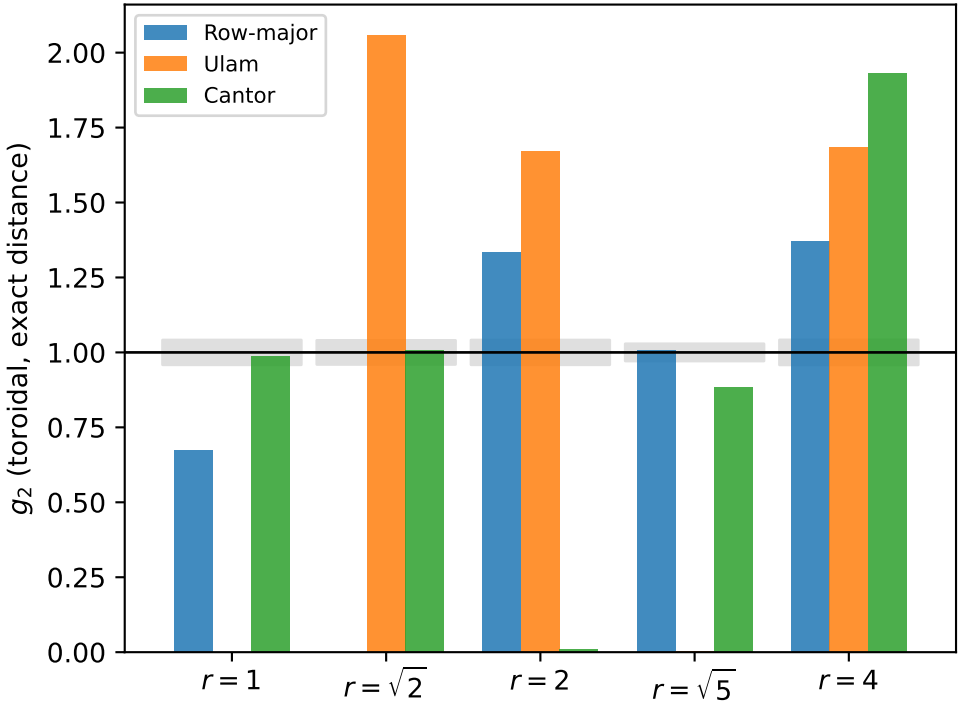
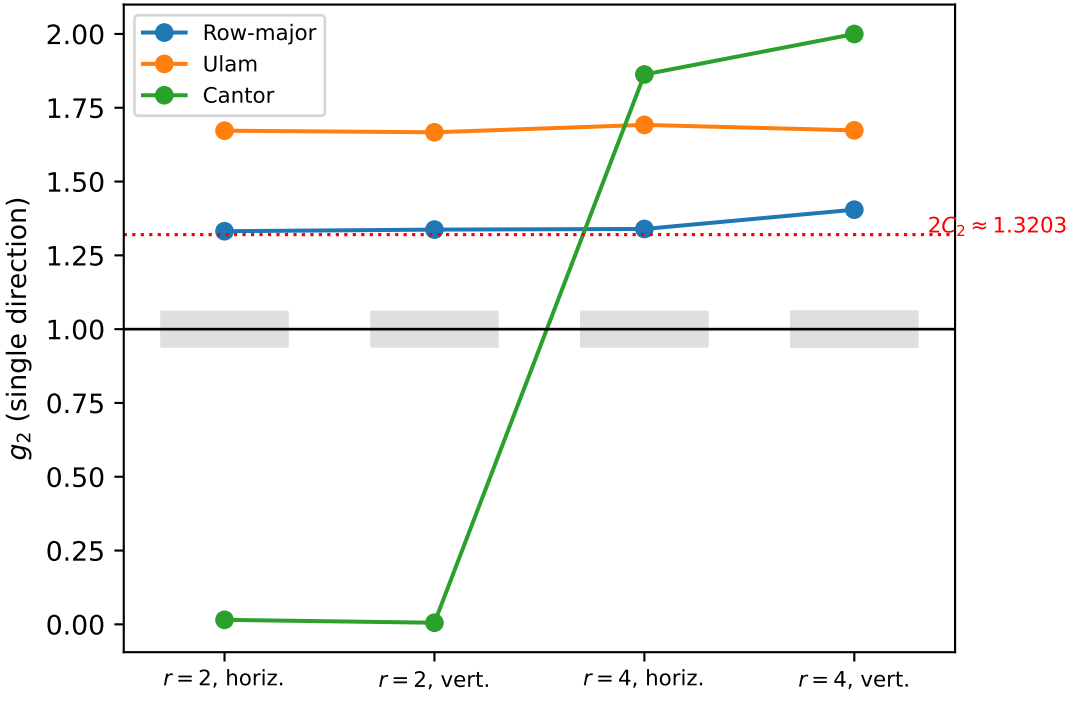

(a) Exact-distance pair correlations vs lattice Bernoulli
Row-major
Ulam
Cantor
g_2 (toroidal, exact distance)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
$r = 1$
$r = \sqrt{2}$
$r = 2$
$r = \sqrt{5}$
$r = 4$


(b) Direction-resolved: twin ($r = 2$) and cousin ($r = 4$) separations
Row-major
Ulam
Cantor
g_2 (single direction)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
$2C_2 \approx 1.3203$
$r = 2$, horiz.
$r = 2$, vert.
$r = 4$, horiz.
$r = 4$, vert.